# Attachment C

## Affidavit of Christopher Pilong on Behalf of PJM Interconnection, L.L.C.

**UNITED STATES OF AMERICA**
**BEFORE THE**
**FEDERAL ENERGY REGULATORY COMMISSION**

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| **PJM Interconnection, L.L.C.** | ) | **Docket No. ER26-____-000** |

**AFFIDAVIT OF CHRISTOPHER PILONG**
**ON BEHALF OF PJM INTERCONNECTION, L.L.C.**

## I. Introduction

1. My name is Christopher Pilong. My business address is 2750 Monroe Boulevard, Audubon, Pennsylvania, 19403. I am Senior Director of Operations Planning for PJM Interconnection, L.L.C. ("PJM"). I am submitting this affidavit on behalf of PJM to discuss when and how PJM would utilize the Interim Resource Adequacy Service ("IRAS") to manage system reliability. Specifically, in this affidavit, I explain how PJM will generally follow its existing processes and procedures in calling for IRAS load reductions and provide some examples of how it will work in practice.

2. I joined PJM in November 2001. Currently, as Senior Director of Operations Planning, I am responsible for the Transmission Operations Department, Generation Department, Transmission Service Department, and Outage Analysis Technologies Department. These departments perform near-term transmission outage analysis and approvals, generation outage analysis and approvals, facilitate annual Black Start Unit commitments, support and develop operational tools, implement and comply with all North American Energy Standards Board business rules regarding Available Flowgate Capability/Available Transfer Capability, perform gas-electric coordination activities, as well as many other tasks to prepare for, and support, real time operations. The teams also perform seasonal reliability assessments, support the various tools and applications utilized in these processes, and coordinate directly with the Market Services Division and System Planning Division.

3. Previously, as Director, Dispatch, I was responsible for the oversight and operation of the Valley Forge and Milford Control Centers. This function included ensuring the reliable operation of the power grid, in accordance with all PJM and North American Electric Reliability Corporation ("NERC") reliability standards pertaining to the functions of Reliability Coordinator, Balancing Authority, and Transmission Operator. In addition, I was responsible for ensuring the efficient economic dispatch of the system under the existing PJM market rules and neighboring Joint Operating Agreements.

4. Prior to that position, I served as a Reliability Engineer and then Manager – Reliability Engineering. As Manager for the Reliability Engineering Group, I managed the group responsible for coordinating day-ahead and real-time operating plans among PJM staff, PJM Transmission Owners, PJM Generation Owners, and our neighboring entities. I performed these functions directly in my earlier role as a Reliability Engineer. I hold a

Bachelor of Science degree in Electrical Engineering from Lehigh University and a Master of Business Administration degree from Villanova University.

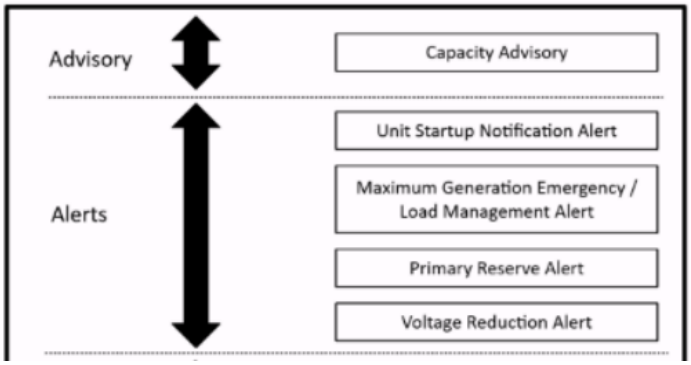
## II. Overview of Practices When Capacity Shortages Give Rise to Emergency Procedures

5. As set forth in section 2.1 of PJM Manual 13, "PJM is responsible for determining and declaring that an Emergency is expected to exist, exists, or has ceased to exist in any part of the PJM [regional transmission organization ("RTO")] or in any other Control Area that is interconnected directly or indirectly with the PJM RTO. [1]."[2]

6. In PJM, there are three general levels of Emergency Procedures for capacity shortages, and an advisory level.

7. Prior to the Operating Day, PJM may issue advisories and/or alerts (see Figure 1, below). The advisory level is general in nature and for elevated awareness "only and may be issued one or more days in advance of the Operating Day."[3] Emergency Procedure Alerts may be "issued one or more days in advance of the Operating Day for elevated awareness and to give time for advanced preparations."[4]

Figure 1: Advisory and Alerts[5]




### A. Steps in Real-Time Prior to Emergency Actions

8. PJM Dispatchers monitor projected and current reserves and determine whether actions are needed to maintain reserves within the required limits. In accordance with PJM Manual 13, before entering into capacity-related emergency actions in real-time, PJM Dispatch will

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[2] Systems Operations Division, *PJM Manual 13: Emergency Operations*, PJM Interconnection, L.L.C., section 2.1 (rev. 98, June 24, 2026), https://www.pjm.com/-/media/DotCom/documents/manuals/m13.pdf [https://perma.cc/HE9Y-BNQK] ("PJM Manual 13").

[3] PJM Manual 13, section 2.3.

[4] PJM Manual 13, section 2.3.

[5] PJM Manual 13, section 2.3.

use available generation and economic demand response, based on economic dispatch, which ensures that Real-Time Locational Marginal Prices reflect actual system conditions.[6]

9. Second, as set forth in PJM Manual 13, if generation and economic demand resources are not sufficient to maintain reserves, PJM Dispatch will then recall energy from its Capacity Resources by curtailing non-firm export transactions, i.e., interchange transactions, in accordance with Part II of the PJM Open Access Transmission Tariff ("Tariff").

10. PJM Dispatch will also have declared a NERC Energy Emergency Alert Level 1 ("EEA1"), as required by NERC Standard EOP-011 Attachment 1, via the Reliability Coordinator Information System and Emergency Procedures webpage.[7]

11. There are three levels of NERC Energy Emergency Alert Levels (EEA1 through EEA3), with an EEA3 being the highest alert level. "EEA1 signals that PJM foresees or is experiencing conditions where all available resources are scheduled to meet firm load, firm transactions, and reserve commitments, and is concerned about sustaining its required Contingency Reserves."[8] "An EEA3 is issued when [PJM] is unable to meet minimum Contingency Reserve Requirements."[9]

12. Third, if system conditions remain constrained and reserves are projected to be short of requirements, assuming PJM's new, yet-to-be-effective transmission service products—Non-Firm Contract Demand Service ("CDS") and Interim Network Integrated Transmission Service ("Interim NITS") are effective, PJM may direct curtailment of Non-Firm CDS and/or Interim NITS transmission service.[10]

13. Fourth, if system conditions remain constrained and reserves are projected to be short of requirements, then PJM will curtail the new IRAS load as outlined in the Examples below.

14. Fifth, if a system emergency develops, then PJM will initiate Steps 1 through Step 10 of the Capacity Emergency procedures as necessary to maintain the reliability of the grid. If Step 10 (Manual Load Dump Action) is reached, this may require firm load to shed pursuant to emergency actions. PJM members are expected to implement all emergency procedures based on the timeliness associated with each specific procedure.[11]

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[6] PJM Manual 13, section 2.3.2.

[7] PJM Manual 13, section 2.3.2.

[8] PJM Manual 13, section 2.3.1.

[9] PJM Manual 13, section 2.3.2.

[10] *PJM Interconnection, L.L.C.*, 195 FERC ¶ 61,209, at P 551 (2026) ("[W]e find PJM's proposal to curtail Interim NITS and [Non-Firm Contract Demand] transmission service prior to deployment of its Pre-Emergency Load Response Program, or in response to a transfer capability shortage as a result of system reliability conditions, to be just and reasonable because it is consistent with how Non-Firm [Point-to-Point Service] is curtailed. . . . [W]e find that PJM's proposal to curtail Interim NITS during capacity or reserve shortages is just and reasonable and consistent with both the nature of the transmission service as a non-firm transmission service for which Eligible Customers do not pay generation capacity charges, and the need to maintain reliability." (footnote omitted)).

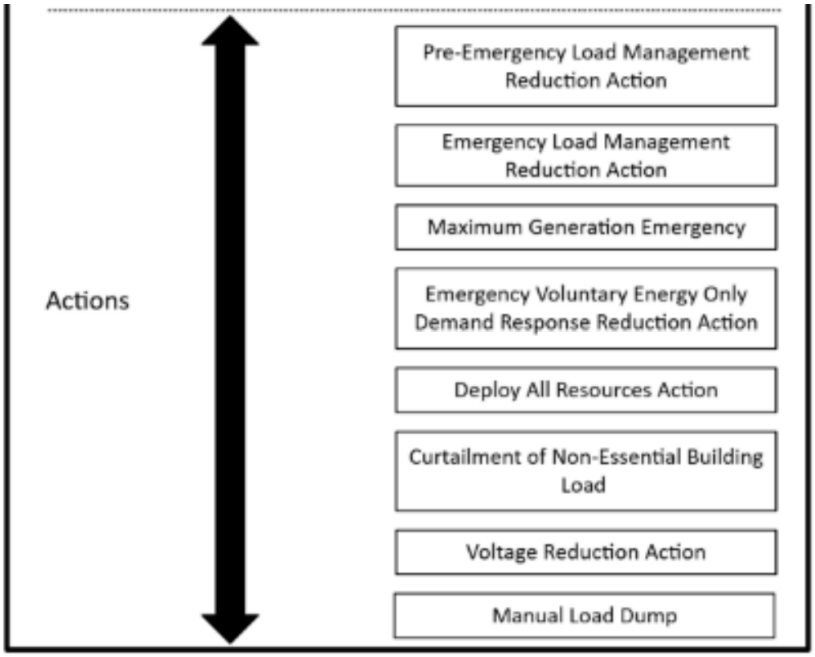
[11] PJM Manual 13, section 2.3.

### B. Steps for Real-Time Emergency Actions Set Forth in PJM Manual 13

15. Importantly, while there is a sequence of escalating emergency action steps set forth in PJM Manual 13, the rapidness of system events influences the timing and order of actions necessary to maintain system reliability. Therefore, PJM Dispatchers "have the flexibility to implement emergency procedures in whatever order is required to ensure system reliability.[12] Likewise, they have the flexibility to exit emergency procedures in a different order than they were implemented."[13]

16. Figure 2, below, excerpted from PJM Manual 13, illustrates the escalating emergency actions that PJM may take and/or require of Members under the status quo rules.[14]

Figure 2.




17. As Figure 2, above, demonstrates, under current Emergency Procedures, the first step when an emergency action is declared is to require Pre-Emergency Load Management Reductions, which essentially require a class of Demand Resources and DER Aggregation Resources to reduce load.

18. At the most extreme, PJM will declare an EEA3, in which PJM may call for a Manual Load Dump (i.e., shedding load taking NITS), a last-resort emergency instruction issued by PJM, triggered during severe capacity deficiencies or critical transmission overloads when all other operational steps have failed.

[12] PJM Manual 13, section 2.3.

[13] PJM Manual 13, section 2.3.

[14] PJM Manual 13, section 2.3.

## III. Addition of Interim Resource Adequacy Service Emergency Procedures to Existing Emergency Procedures in PJM Manual 13

19. New Large Loads are developing more quickly than generation and transmission can be built. As a result, this may impact the ability for PJM to maintain a reliable system and continue to meet the demand for existing customers and maintain reserve requirements during times of peak demand and/or significant generator outages. Therefore, a new operational tool is needed to ensure the current level of reliability for our customers is maintained.

20. IRAS, as proposed, is an additional load reduction service that PJM may deploy in real-time, starting with the 2027/2028 Delivery Year. As part of this filing, PJM is proposing a new term, "Interim Resource Adequacy Service Action," to mean "an Operating Instruction, as defined by NERC, issued from the Office of the Interconnection to reduce load prior to calling on Pre-Emergency Load Management Reductions after determining that available economic real-time supply may not be sufficient to maintain reserves to meet applicable Reserve Requirements and/or Interconnection Reliability Operating Limits ("IROL") within prescribed limits and prevent cascading outages."[15]

21. The IRAS framework is intended to operate as both a forward-looking reliability assurance mechanism and an operational tool. The identification of an IRAS requirement for each Electric Distributor's zone/area establishes the reliability obligation applicable to affected load, while PJM's real-time operating processes determine when and how that obligation is activated based on actual system conditions.

22. The IRAS framework therefore does not require PJM to reduce affected load whenever an IRAS requirement exists. Rather, the IRAS requirement establishes an operational reliability resource available to PJM, which can be called upon when actual system conditions indicate that the reliability need addressed by the IRAS requirement is materializing.

23. PJM will incorporate IRAS-designated load, that PJM will determine under the IRAS rules, into its operating models, operating plans, and real-time reliability assessments. PJM's existing real-time Energy Management System and contingency-analysis processes provide the platform for evaluating system conditions and identifying potential reliability violations.

24. If the proposed IRAS framework is accepted by the Commission, PJM Manual 13 will be revised to establish the specific details of how PJM will deploy IRAS through new Emergency Procedures, i.e., Emergency Alerts, Actions, Advisories, and/or Warnings. As is the norm, PJM will work with stakeholders to revise PJM Manual 13 to establish specific language for IRAS Emergency Procedures after the IRAS proposal has been filed with the

[15] Tariff, Definitions – Interconnection Reliability Operating Limits; *Glossary Terms*, North American Electric Reliability Corporation, https://www.nerc.com/glossary-of-terms?alpha=I [https://perma.cc/TZ9W-MZ4K] (last visited Aug. 11, 2026) (defining Interconnection Reliability Operating Limit as "[a] System Operating Limit that, if violated, could lead to instability, uncontrolled separation, or Cascading outages that adversely impact the reliability of the Bulk Electric System.").

Commission. Any revisions to PJM Manual 13 to reflect IRAS will not become effective unless the Commission accepts the IRAS framework.

25. That said, I will provide, below, a general overview of how IRAS, as a new operational tool, will be deployed. PJM expects that existing Emergency Procedures set forth in PJM Manual 13 will not change except to add IRAS as another potential tool at PJM's disposal that PJM may deploy in efforts to address a potential emergency prior to Pre-Emergency Load Management Reduction Action.

26. In day-ahead, Interim Resource Adequacy Service Alerts may precede a Maximum Generation Alert ("Max Gen Alert"). Thus, IRAS provides a new, lower-level Emergency Procedure Alert that PJM Dispatch may issue prior to a Max Gen Alert. In real-time, PJM Dispatch will deploy Interim Resource Adequacy Service Actions before Pre-Emergency Load Management Reduction Actions. Thus, in real-time, IRAS provides a new step, at PJM's disposal, prior to impacting existing customers. If the Interim Resource Adequacy Service Actions do not address the emergency, PJM Dispatchers would then take necessary Pre-Emergency Load Management Reduction Actions and declare escalating Emergency Procedures as necessary.

27. In the next subsections of my affidavit, I explain how a PJM Dispatcher may deploy IRAS. In Example 1, below, I describe how PJM issues an Interim Resource Adequacy Service Alert in the day-ahead. In Example 2, I describe a scenario when, in real-time, PJM experiences an RTO-wide reliability and/or capacity shortfall event and responds by deploying IRAS. In Example 3, in real-time, PJM experiences an IROL and responds by deploying IRAS.

28. It is important to note that prior to the start of each Delivery Year, the megawatts ("MW") quantity of IRAS that will be required to respond to an Interim Resource Adequacy Service Alert will be determined by PJM for each Electric Distributor zone/area. This value will be utilized by PJM Dispatch in their system projections, alerts and actions in the examples below (e.g., PJM may determine, for example, that prior to the Delivery Year, there may be 4 gigawatts of IRAS available across the RTO, with smaller amounts assigned to an Electric Distributor zone/area for real-time curtailment if an Interim Resource Adequacy Service Action is issued).

### A. Example 1—Day-Ahead Interim Resource Adequacy Service Alert

29. As PJM operators look ahead to the next Operating Day, they periodically evaluate forecasted demand and scheduled supply. They also evaluate, among many other things, expected weather conditions to determine, e.g., wind and solar resource availability, temperature, etc. PJM schedules reserves on a day-ahead basis to address such contingencies and variabilities.

30. In this example, in day-ahead, PJM operators reasonably expect that insufficient supply will be available to meet expected real-time demand and reserves for the next Operating Day. PJM's operators will have available to them, on a daily basis, the expected aggregate amount of Interim Resource Adequacy Service MW that can be reduced at the Electric

Distributor level. If reduction of some/all of the IRAS load may be necessary to meet real-time demand and reserve requirements, PJM Dispatchers may elect to issue an Interim Resource Adequacy Service Alert in day-ahead.

31. PJM issues an Interim Resource Adequacy Service Alert before and/or concurrently with a Max Gen Alert/Load Management Alert. The Interim Resource Adequacy Service Alert puts all Eligible New Large Loads on notice that PJM could request reductions in load in quantities equal to or lesser than each Electric Distributor zone/area's assigned IRAS amounts during the next Operating Day. However, under an alert, no IRAS reduction is actually requested or implemented. This is simply a notice to prepare and will be issued to provide maximum notice possible.

32. In this example, PJM also elects to issue a Max Gen Alert/Load Management Alert in day-ahead.

33. As explained in section 2.3.1 of PJM Manual 13, "The intent of the alert(s) is to keep all affected system personnel aware of the forecasted and/or actual status of the PJM RTO. All alerts and cancellation thereof are broadcast on the ALLCALL system and posted to selected PJM websites to assure that all members receive the same information. Alerts are issued in advance of a scheduled load period to allow sufficient time for members to prepare for anticipated initial capacity shortages."[16]

34. For Examples 2-3, where PJM issues an Interim Resource Adequacy Service Action in real-time, it can be assumed that PJM Dispatch would have already issued an Interim Resource Adequacy Service Alert. Thus, in real-time, PJM's members are already aware that potential Emergency Procedure actions may take place in real-time. However, it should be noted that not all conditions can be forecasted, and it is not a requirement that an Interim Resource Adequacy Service Alert be issued prior to an Interim Resource Adequacy Service Action. Therefore, it is possible to have no day-ahead alerts, but nevertheless have an Interim Resource Adequacy Service Action or other Actions issued during the Operating Day.

### B. Example 2—Real-Time Interim Resource Adequacy Service Action, RTO-Wide Emergency

35. In real-time, PJM faces an RTO-wide system emergency. Prior to directing an Interim Resource Adequacy Service Action, however, PJM has several other ways to address RTO-wide reliability concerns.

36. First, in keeping with the practice in place today and set forth in PJM Manual 13, PJM Dispatchers deploy available generation and economic demand resources, in economic dispatch order while maintaining required reserves. Even with all available economic resources dispatched, PJM's operators may still face a potential reliability issue and capacity shortfall in real-time.

[16] PJM Manual 13, section 2.3.1.

37. Second, in keeping with the practice in place today and as set forth in PJM Manual 13, if issues persist or are projected to deteriorate, PJM operators may require non-firm interchange transaction curtailment, as needed.

38. PJM Dispatchers will have already issued an EEA1. After the first and second steps taken by PJM Dispatchers, PJM remains under an EEA1.

39. Third, assuming for purposes of this example that the yet-to-be-effective Interim NITS and Non-Firm CDS transmission services are now in place, PJM may curtail Non-Firm CDS, followed by Interim NITS.

40. Fourth, after all non-firm transmission services are curtailed, PJM Dispatchers would then issue the new Interim Resource Adequacy Service Action prior to deployment of a Pre-Emergency Load Management Reduction Action.

41. PJM intends to issue an Interim Resource Adequacy Service Action through the status quo notification processes set forth in PJM Manual 13.[17] Pursuant to the Interim Resource Adequacy Service Action, PJM prioritizes load reduction from those Electric Distributor zones/areas where New Large Loads have not brought their own new capacity and do not have Allocated Reliability Backstop Procurement Unforced Capacity Megawatts ("RBP UCAP MW") sufficient to cover their capacity needs. PJM's operators will direct the IRAS reduction to Electric Distributor zones/areas as needed based on system conditions.

42. More specifically, through an Interim Resource Adequacy Service Action, PJM Dispatchers will direct Electric Distributors to implement reductions in load up to, and capped at, the pre-Delivery Year IRAS obligations they were assigned. Importantly, PJM will not direct specific New Large Loads to reduce load, even though Electric Distributors—in coordination, where applicable, with relevant states, Load Serving Entities, and New Large Loads—may issue more specific directives to specific retail customers.[18] PJM will only direct load reduction of an aggregate MW IRAS within the zone or area.

43. In this example, the Interim Resource Adequacy Service Action is deployed in response to a Region-wide emergency. As such, based on the maximum IRAS amounts assigned to each Electric Distributor zone/area, PJM Dispatchers call for some or all Electric Distributors to reduce load across the PJM RTO Region in the manner they see fit to

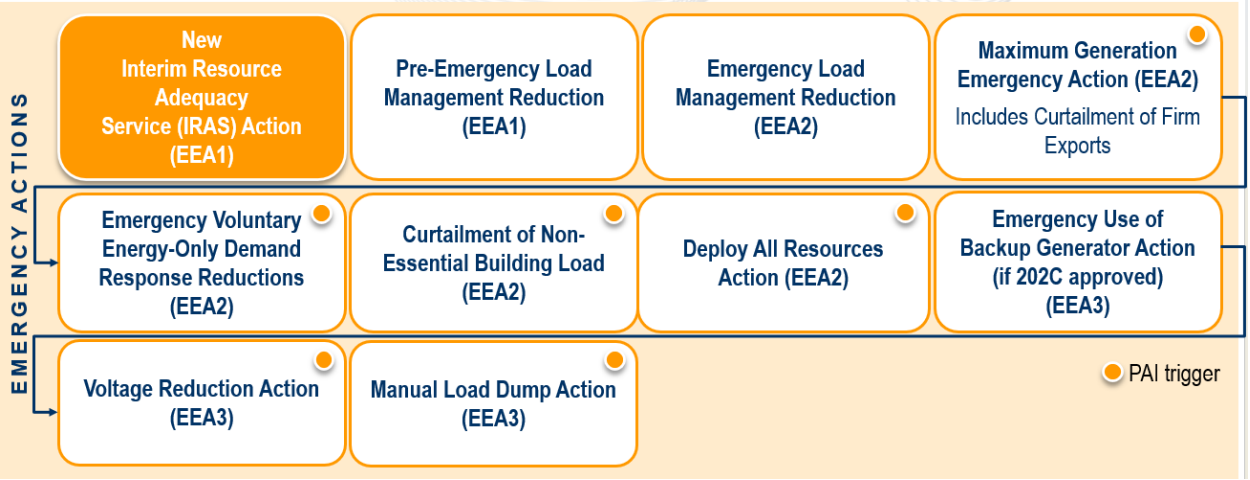
[17] System Operations Division, *PJM Manual 37: Reliability Coordination*, PJM Interconnection, L.L.C., section 3.2 (rev. 23 Apr. 22, 2026), https://www.pjm.com/-/media/DotCom/documents/manuals/m37.pdf [https://perma.cc/FY3P-8SCM].

[18] As PJM Manual 13 notes, PJM members have Member Load shed plans, which, among other things, "recognize priority and critical load including: Essential health and public safety facilities such as hospitals, police, fire facilities, 911 facilities, wastewater treatment facilities; Facilities providing electric service to facilities associated with Bulk Electric System including off-site power to generating stations, substation light and power; Critical natural gas infrastructure used to supply gas pipeline pumping plants, processing and production facilities; and Telecommunication facilities." PJM Manual 13, section 2.3.2.

address the system conditions. As directed by PJM, those Electric Distributors then reduce load in accordance with PJM's emergency action.

44. Figure 3, below, shows how an Interim Resource Adequacy Service Action fits into PJM's existing Emergency Procedures framework.

**Figure 3: PJM Emergency Procedures with Interim Resource Adequacy Service (IRAS)**




45. As shown in Figure 3, the Interim Resource Adequacy Service Action precedes Pre-Emergency Load Management Reductions in PJM's order of operations to maintain reliability.

46. In this example, the EEA1 continues, and PJM Dispatchers then proceed to Pre-Emergency Load Management Reductions, in keeping with the status quo steps set forth in PJM Manual 13.[19]

47. The Pre-Emergency Load Management Reductions address the RTO-wide emergency. PJM Dispatch cancels the Pre-Emergency Load Management Reductions and Interim Resource Adequacy Service Actions, when appropriate.

### C. Example 3—Real-Time Interim Resource Adequacy Service Action, Potential Interconnection Reliability Operating Limit Violation

48. In my next real-time example, there is a potential IROL violation. Certain facilities, known as IROL facilities, can significantly impact system reliability if loaded above a designated limit.[20] If an IROL facility's designated limits are exceeded, this has the potential to result

[19] System Operations Division, *PJM Manual 37: Reliability Coordination*, PJM Interconnection, L.L.C., section 3.2, (rev. 23 Apr. 22, 2026), https://www.pjm.com/-/media/DotCom/documents/manuals/m37.pdf [https://perma.cc/FY3P-8SCM].

[20] PJM Manual 13, section 5.5 (including identification of IROL facilities).

in wide-area voltage collapse.[21] Therefore, PJM Dispatch must quickly act to mitigate IROL violations.

49. An IROL violation, unlike a PJM RTO-wide emergency, occurs when one or more IROL facilities between zones become overloaded or is otherwise experiencing zonal capacity shortages. In my example, PJM's response to a potential IROL violation includes issuance of an Interim Resource Adequacy Service Action.

50. PJM monitors potential IROL violations via PJM Energy Management System, which automatically identifies zones and loads that are at risk of triggering IROL facilities to exceed their limits.

51. In the event of a potential IROL violation, PJM engages in similar steps and notifications in real-time described in Example 2, above, but does so in a more targeted manner. For example, the response to the potential IROL violation may not require curtailment of all non-firm exports to the extent the exports do not exacerbate the potential IROL violation.

52. In this case, for example, let's say the Dominion Zone is experiencing a transmission capacity scarcity event resulting from a potential IROL violation due to overloading of an IROL facility in West Virginia, e.g., AP South. In response, it makes little sense for PJM Dispatchers to curtail non-firm exports to the West or North of the PJM system. However, PJM may require curtailment of non-firm exports between PJM and areas to the South that are flowing power across the AP South IROL Reactive Interface to mediate the potential IROL violation.

53. Similarly, when PJM Dispatchers issue an Interim Resource Adequacy Service Action in response to the potential IROL violation, it is a more targeted Interim Resource Adequacy Service Action than the pro-rata load reductions across multiple Electric Distributor zones/areas issued in Example 2, above. If load in a particular Electric Distributor zone/area is not negatively impacting transmission flows into the Dominion Zone, load reduction in that zone/area may not have any effect on the Dominion Zone.

54. The Interim Resource Adequacy Service Action, in the event of an IROL violation, will require similar steps in real-time as an Interim Resource Adequacy Service Action across the entire PJM RTO Region, except that PJM Dispatchers will only deploy load reductions up to the maximum IRAS amounts PJM determined for those affected Electric Distributors' zones/areas needed to alleviate the transmission capacity scarcity event within that specific zone/area.

55. Through distribution factor modeling from its Energy Management System, PJM will identify the Electric Distributors' zone(s)/area(s) in which an Interim Resource Adequacy Service Action will have the greatest impact in alleviating the IROL violation. PJM will then direct those relevant Electric Distributors to reduce load. The Electric Distributor, in

[21] System Operations Division, *PJM Manual 37: Reliability Coordination*, PJM Interconnection, L.L.C., section 3.2, (rev. 23Apr. 22, 2026), https://www.pjm.com/-/media/DotCom/documents/manuals/m37.pdf [https://perma.cc/FY3P-8SCM].

coordination with applicable states, Load Serving Entities, and Large Loads, will, in turn, direct specific loads to reduce in response to the Interim Resource Adequacy Service Action.

56. This concludes my affidavit.

**UNITED STATES OF AMERICA**
**BEFORE THE**
**FEDERAL ENERGY REGULATORY COMMISSION**

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| **PJM Interconnection, L.L.C.** | **)** | **Docket No. ER26-____-000** |

## VERIFICATION

I, **Christopher Pilong**, state, under penalty of perjury, that I am the Christopher Pilong referred to in the foregoing document entitled, "Affidavit of Christopher Pilong on Behalf of PJM Interconnection, L.L.C.," that I have read the same and am familiar with the contents thereof, and that the facts set forth therein are true and correct to the best of my knowledge, information, and belief.

/s/ *Christopher Pilong*
Christopher Pilong